

Supplementary Table 1: Complete parameter list of the NED model

Parameter	Module	Species / Process	Value in notebook	Typical range	Status	Basis	Section
<i>NICD</i> module							
Vmni	NICD_reg	<i>NICD</i> max production (Hp)	25	—	fixed	phenomenological	§1.1
hMni	NICD_reg	<i>NICD</i> activation $K_{0.5}$	2000	—	fixed	phenomenological	§1.1
Kdni	NICD_reg	<i>NICD</i> degradation rate	0.25	0.1–0.5	scanned	experimentally constrained	§1.1
tmDN	CC3D → <i>NICD</i>	Trans <i>DLL4</i> – <i>NOTCH1</i> interactions	computed each MCS	0–5000	dynamic	structural (coupling)	EC_Cor 235
cmDN	cis_interactions	Productive cis <i>DLL4</i> – <i>NOTCH1</i>	computed	0–5000	dynamic	structural (coupling)	§2.4
Notch / ligand expression							
bN1	ligands_receptors_reg	Basal <i>NOTCH1</i> production	250	—	fixed	phenomenological	§2.1
Kdn1	ligands_receptors_reg	<i>NOTCH1</i> degradation	0.05	—	fixed	phenomenological/literature constrained	§2.1
hMn1	ligands_receptors_reg	<i>NOTCH1</i> inducibility $K_{0.5}$	50	—	fixed	phenomenological	§2.1
N1	ligands_receptors_reg	<i>NOTCH1</i> level	dynamic	0–5000	dynamic	structural (state)	§2.1
bD4	ligands_receptors_reg	Basal <i>DLL4</i> production	250	—	fixed	phenomenological	§2.2
hMd4	ligands_receptors_reg	<i>DLL4</i> repression $K_{0.5}$	50	—	fixed	phenomenological	§2.2
Kdd4	ligands_receptors_reg	<i>DLL4</i> degradation	0.05	—	fixed	phenomenological/literature constrained	§2.2
dD4	CC3D → Antimony	Engaged <i>DLL4</i> loss term	computed each MCS	0– <i>D4</i> -dependent	dynamic	structural (coupling)	EC_Cor 235
D4	ligands_receptors_reg	<i>DLL4</i> level	dynamic	0–5000	dynamic	structural (state)	§2.2

VmJ1	ligands_receptors_reg	Max <i>Jagged1</i> induction	250	—	fixed	phenomenological	§2.3
hMj1	ligands_receptors_reg	<i>Jagged1</i> induction $K_{0.5}$	ps_ja (≈ 50)	25–100	scanned	phenomenological	§2.3
bJ1	ligands_receptors_reg	Basal <i>Jagged1</i> production	0	—	fixed	phenomenological	§2.3
Kdj1	ligands_receptors_reg	<i>Jagged1</i> degradation	0.05	—	fixed	phenomenological/literature constrained	§2.3
dJ1	CC3D $\rightarrow$ Antimony	Engaged <i>Jagged1</i> loss term	computed each MCS	0– <i>J1</i> -dependent	dynamic	structural (coupling)	EC_Cor 235
<i>J1</i>	ligands_receptors_reg	<i>Jagged1</i> level	dynamic	0–5000	dynamic	structural (state)	§2.3
Trans interactions							
tmJN	CC3D $\rightarrow$ <i>NICD</i>	Trans <i>Jagged1</i> – <i>NOTCH1</i>	computed each MCS	0–5000	dynamic	structural (coupling)	EC_Cor 235
Rd4	CC3D	Reacted <i>DLL4</i>	computed	0–5000	dynamic	structural (coupling)	EC_Cor 235
Rj1	CC3D	Reacted <i>Jagged1</i>	computed	0–5000	dynamic	structural (coupling)	EC_Cor 235
Cis interactions							
cJN	cis_interactions	Non-productive <i>Jagged1</i> –Notch cis	mass-action	n/a	fixed law	structural	§2.4
cdn	cis_interactions	Productive <i>DLL4</i> –Notch cis	mass-action	n/a	fixed law	structural	§2.4
cadn	cis_interactions	Active cis <i>DLL4</i> –Notch	derived	0–5000	dynamic	structural (state)	§2.4
Competition parameters							
kpDJ	CC3D	<i>DLL4</i> / <i>Jagged1</i> partitioning	computed each MCS	0–1	dynamic	phenomenological control	EC_Cor 235
KJC	ligands_receptors_reg	<i>Jagged1</i> affinity modifier	1.0	0.5–2	fixed	phenomenological control	EC_Cor 235
<i>HES1</i> – no autoregulation							
VmH1	HES_no_auto	Max <i>HES1</i> production	10	—	fixed	phenomenological	§3.1
hMh1	HES_no_auto	<i>NICD</i> $\rightarrow$ <i>HES1</i> $K_{0.5}$	35	—	fixed	phenomenological	§3.1

Kdh1	HES_no_auto	<i>HES1</i> degradation	0.09	—	fixed	experimentally constrained	§3.1
bH1	HES_no_auto	Basal <i>HES1</i> production	0.0	—	fixed	phenomenological	§3.1
<i>HES1</i> – autoregulation							
vm_prna	HES_auto	Max pre- <i>mRNA</i> transcription	0.1	0.05–0.2	scanned	phenomenological	§3.2.2
K	HES_auto	Shared $K_{0.5}$ (Hill/MM)	0.5	0.3–0.7	scanned	phenomenological	§3.2.2
hh1	HES_auto	<i>HES1</i> autorepression Hill	8	6–10	fixed	structural required	§3.2.2
kmrp	HES_auto	<i>mRNA</i> processing rate	0.1	0.05–0.2	scanned	structural required	§3.2.2
kmrd	HES_auto	<i>mRNA</i> degradation	0.05	0.02–0.1	scanned	structural required	§3.2.2
kphp	HES_auto	Translation rate	0.1	0.05–0.2	scanned	structural required	§3.2.2
khp	HES_auto	Protein maturation	0.1	0.05–0.2	scanned	structural required	§3.2.2
khd	HES_auto	Active <i>HES1</i> degradation	0.075	0.05–0.15	scanned	experimentally constrained	§3.2.2
<i>NICD</i>	HES_auto	<i>NICD</i> input (scaled)	0–1	0–1	controlled	experimental driver	§3.2.3
Utility / kinetics							
Hp()	utility	Positive Hill	—	—	structural	—	§0 (utili
Hc()	utility	Competitive Hill	—	—	structural	—	§0 (utili
Ma()	utility	Mass action	—	—	structural	—	§0 (utili
MM()	utility	Michaelis–Menten	—	—	structural	—	§0 (utili

Notes:

- *NICD* amplitude is calculated by V_{mni}/K_{dni} (~100 with 25/0.25); scanning *Kdni* changes the timescale while preserving this range.
- *N1*, *D4*, and *J1* basals/decays place steady states in the 10^3 – 10^4 regime (lumped units), consistent with Boareto et al PNAS 2015.
- Time unit in the Antimony layer is 1 min; characteristic times (e.g., $1/K_{dn1}$, $1/K_{dd4}$, $1/K_{dj1}$) are in real minutes.
- Typical ranges are provided where parameters are scanned or vary during simulations; fixed entries list “—”.